Choosing the Form That Shows the Structure
Answer Key

Algebra 1   Expressions and polynomial operations

Quick Answers

1. $x^2 + 8x + 15$
2. $6x^2 - 14x$
3. $(x - 5)(x - 4)$
4. $(x - 7)(x + 7)$
5. 0
6. $4x^2 - 12x + 9$
7. $(x + 4)(3x - 1)$
8. $12x$
9. $x^3 + 8$
10. $2x(x - 3)(x + 3)$
11. 9991
12. $(x - 3)(x - 2)(x + 2)(x + 3)$, 336

Curriculum

Level: Algebra 1   Expressions and polynomial operations
HSA-APR.A.1 – *problems 1, 2, 6, 8, 9*
HSA-SSE.B.3 – *problems 3, 4, 5, 7, 10, 11, 12*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

5. *If they got 8: added the two factor values instead of multiplying them*
11. *If they got 9409: used the square-of-a-binomial pattern instead of the difference of squares*

1. Expand $(x + 5)(x + 3)$.

Step 1. The question asks for a sum of terms, so the product has to be distributed — every term of the first factor times every term of the second.
Step 2. $x \cdot x = x^2$, then $x \cdot 3 + 5 \cdot x = 8x$, then $5 \cdot 3 = 15$.
Step 3. Collecting the terms gives $\boxed{x^2 + 8x + 15}$.
What the form shows: the constant 15 is the value of the expression at $x = 0$, which the factored form only hides.

2. Expand $2x(3x - 7)$.

Step 1. A monomial multiplying a binomial needs the distributive property once; there is no FOIL here.
Step 2. $2x \cdot 3x = 6x^2$ and $2x \cdot (-7) = -14x$.
Step 3. The expanded form is $\boxed{6x^2 - 14x}$.
Common wrong answer: $6x^2 - 7$ — the $2x$ was distributed only to the first term.

3. Factor $x^2 - 9x + 20$.
Step 1. The question asks what makes the expression zero, so the goal is a product: find two numbers whose product is 20 and whose sum is -9.
Step 2. Both numbers must be negative because the product is positive and the sum is negative. The pair is -4 and -5.
Step 3. So $x^2 - 9x + 20 = \boxed{(x-4)(x-5)}$.
What the form shows: the expression is zero exactly at $x = 4$ and $x = 5$, which is unreadable in expanded form.

4. Factor $x^2 - 49$.
Step 1. There is no middle term and both pieces are perfect squares, which is the signature of the difference-of-squares pattern $A^2 - B^2 = (A - B)(A + B)$.
Step 2. Here $A = x$ and $B = 7$, since $49 = 7^2$.
Step 3. Therefore $x^2 - 49 = \boxed{(x-7)(x+7)}$.
Listen for: "it does not factor because there is no middle term." The missing middle term is the reason it *does.*

5. Evaluate $(x - 6)(x + 2)$ at $x = 6$ without expanding.
Step 1. The expression is already a product, and a product is zero as soon as one factor is zero — so check the factors before doing any arithmetic.
Step 2. At $x = 6$ the first factor is $6 - 6 = 0$. The second factor is $6 + 2 = 8$, but it no longer matters.
Step 3. The value is $0 \cdot 8$, so the answer is $\boxed{0}$.
Which form told you: the factored one. Expanded, this is $x^2 - 4x - 12 = 36 - 24 - 12$, which is the same answer after three more steps.
Common wrong answer: 8 — the two factor values were added instead of multiplied.

6. Expand $(2x - 3)^2$.
Step 1. A square of a binomial is that binomial times itself, so distribute rather than squaring the two pieces separately.
Step 2. $(2x)^2 = 4x^2$; the two cross terms are each $2x \cdot (-3) = -6x$, giving $-12x$; and $(-3)^2 = 9$.
Step 3. The expanded form is $\boxed{4x^2 - 12x + 9}$.
Common wrong answer: $4x^2 + 9$ — the two cross terms were dropped.

7. Factor $3x^2 + 11x - 4$.
Step 1. The leading coefficient is not 1, so the pair of numbers has to multiply to $3 \cdot (-4) = -12$ and add to 11.
Step 2. That pair is 12 and -1. Split the middle term: $3x^2 + 12x - x - 4$.
Step 3. Group: $3x(x + 4) - 1(x + 4)$, so the common factor is $x + 4$ and $3x^2 + 11x - 4 = \boxed{(3x-1)(x+4)}$.
Check: $(3x - 1)(x + 4) = 3x^2 + 12x - x - 4 = 3x^2 + 11x - 4$.

8. Simplify $(x+3)^2 - (x-3)^2$.
Step 1. Nothing cancels while the squares are unexpanded, so this is a case where the expanded form is the useful one.
Step 2. $(x+3)^2 = x^2 + 6x + 9$ and $(x-3)^2 = x^2 - 6x + 9$.
Step 3. Subtracting, the x^2 terms and the 9s cancel and the middle terms add: $6x - (-6x)$, so the result is $\boxed{12x}$.
What the form shows: the difference of these two squares is a multiple of x — it is 0 at $x = 0$ and grows in a straight line, neither of which is visible before expanding.

9. Expand $(x+2)(x^2 - 2x + 4)$.
Step 1. Each of the two terms in the first factor multiplies all three terms in the second, so expect six products before collecting.
Step 2. $x(x^2 - 2x + 4) = x^3 - 2x^2 + 4x$ and $2(x^2 - 2x + 4) = 2x^2 - 4x + 8$.
Step 3. Adding, $-2x^2 + 2x^2 = 0$ and $4x - 4x = 0$, leaving $\boxed{x^3 + 8}$.
What the form shows: the product collapses to a sum of cubes, $x^3 + 2^3$. Recognising that pattern backwards is how the factored form is found in the first place.

10. Factor $2x^3 - 18x$.
Step 1. "Completely" means the greatest common factor comes out first; only then is the pattern in what is left visible.
Step 2. The GCF of $2x^3$ and $18x$ is $2x$, so $2x^3 - 18x = 2x(x^2 - 9)$.
Step 3. Now $x^2 - 9$ is a difference of squares, giving $\boxed{2x(x-3)(x+3)}$.
Common wrong answer: $2x(x-3)^2$ — $x^2 - 9$ was read as a perfect square. The zeros are $x = 0$, $x = 3$ and $x = -3$.

11. Evaluate $(m-3)(m+3)$ at $m = 100$.
Step 1. Multiplying 97 by 103 by hand is slow, so look at the structure first: this is a difference of squares waiting to be expanded.
Step 2. $(m-3)(m+3) = m^2 - 9$, and at $m = 100$ that is $10000 - 9$.
Step 3. The value is $\boxed{9991}$.
Which form was chosen: the expanded one — here expanding replaces a two-digit multiplication with a subtraction.
Common wrong answer: 9409 — the square-of-a-binomial pattern was used instead of the difference of squares.

12. Synthesis: $x^4 - 13x^2 + 36$.
Step 1 (a). Only even powers appear, so treat $u = x^2$: the expression is $u^2 - 13u + 36$, an ordinary quadratic.
Step 2 (a). Two numbers multiplying to 36 and adding to -13 are -4 and -9, so $u^2 - 13u + 36 = (u-4)(u-9) = (x^2-4)(x^2-9)$.
Step 3 (a). Each bracket is a difference of squares, so the complete factorisation is $\boxed{(x-2)(x+2)(x-3)(x+3)}$.
Step 4 (b). Substituting $x = 5$ into the factored form gives $(3)(7)(2)(8)$, four small products instead of a fourth power.
Step 5 (b). $3 \cdot 7 = 21$, $2 \cdot 8 = 16$, and $21 \cdot 16$ is $\boxed{336}$.
Check: the original form gives $625 - 325 + 36 = 336$ as well — the same number, with a fourth power to compute.